



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

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SECP3213 BUSINESS INTELLIGENCE

SECTION 01

PROJECT PHASE 2 DATA WORKFLOW

Analysing the Impact of Macroeconomic Indicators and Financial

Performance on Malaysian Banking Stocks

LECTURER: TS. DR. NOR AZIZAH ALI

Student Name	Matric No.
LAM YOKE YU	A23CS0233
TAN YI YA	A23CS0187
TEH RU QIAN	A23CS0191

1. Data Ingestion & Transformation

The datasets were ingested and cleaned, and the datasets were gradually combined as the following figure.

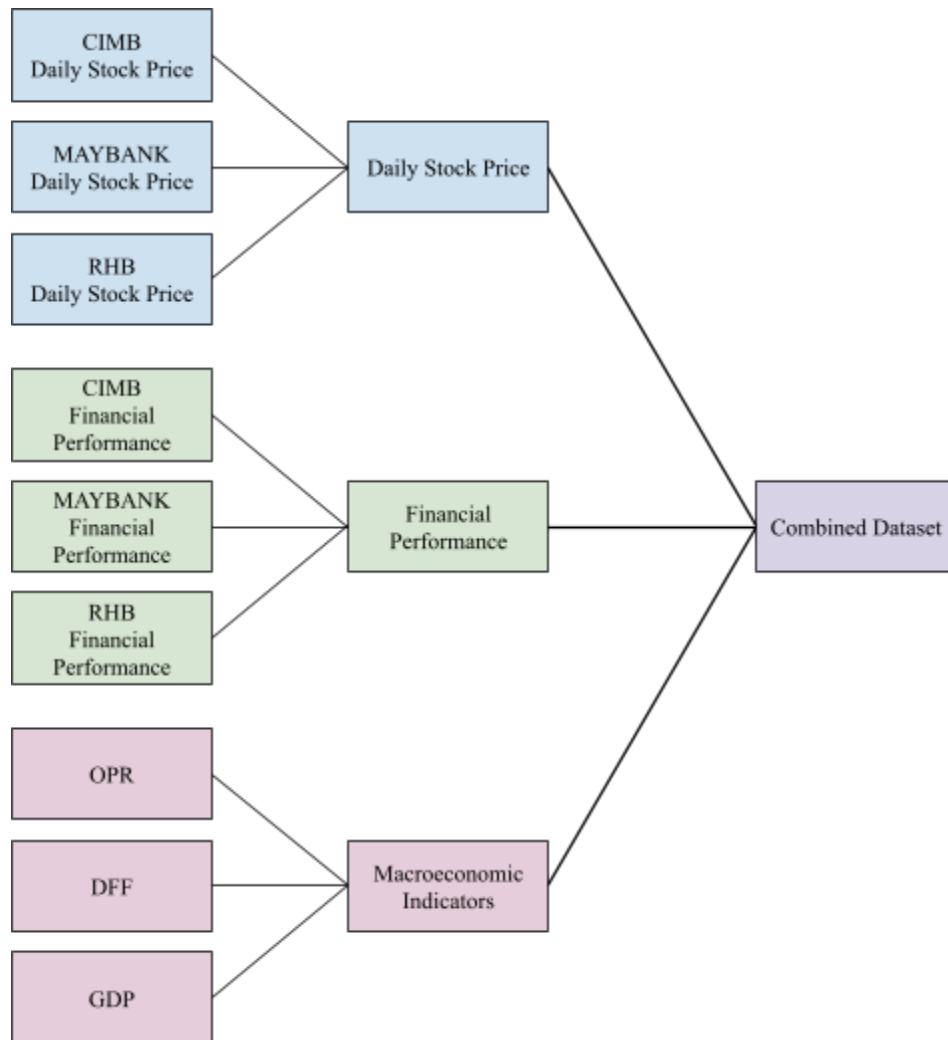


Figure 1 Integration Procedure

1.1 Daily Stock Price

Figure 2 illustrates the workflow used to prepare the daily stock price data for CIMB, MAYBANK and RHBBANK. Historical stock price data from 1 January 2014 to 29 May 2026 were downloaded from Investing.com and imported into Alteryx.

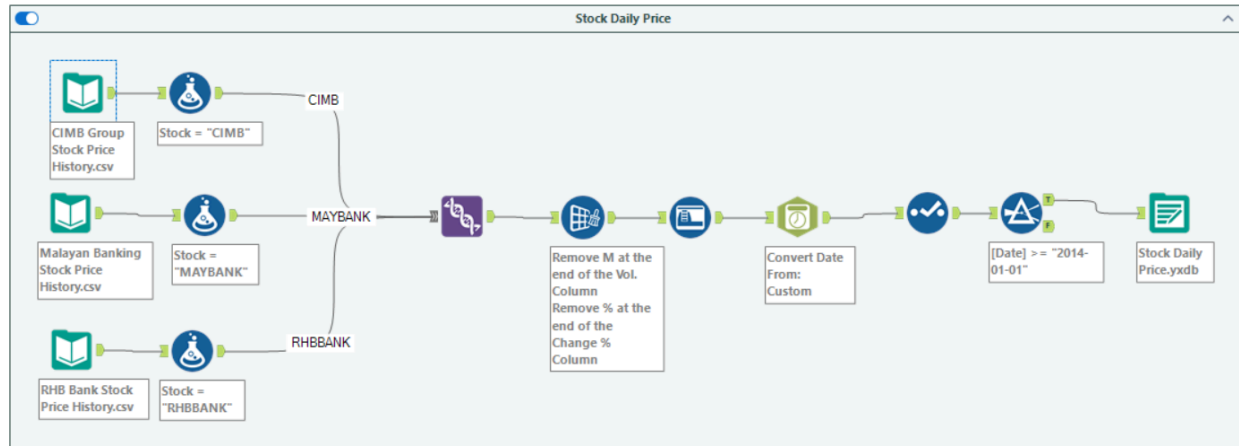


Figure 2 Daily Stock Price Cleaning and Integration Workflow

Tools	Justification
Input Data	Imported CSV files for daily stock price.
Formula	Added “Stock” column to identify each dataset.
Union	Combined the three stock datasets.
Data Cleanse Pro	Cleaned data by removing special characters, letters, and unit symbols in Vol. and Change %.
Auto Field	Converted fields (Price, Open, High, Low, Vol., Change %) to appropriate data types.
DateTime	Standardised the date format.
Select and Filter	Selected and renamed relevant columns and retained data from 2014 onwards.
Output Data	Saved as Stock Daily Price.yxdb.

One of the minor challenges was the auto field not being able to identify the numeric columns of Vol. and Change % due to the suffix characters in the values. The problem was resolved by removing the special characters from the values.

1.2 Financial Performance

Figure 3 shows the workflow used to prepare the financial performance data of CIMB, MAYBANK and RHBBANK. The data were manually compiled from KLSE Screener and stored in Excel format before being imported into Alteryx.

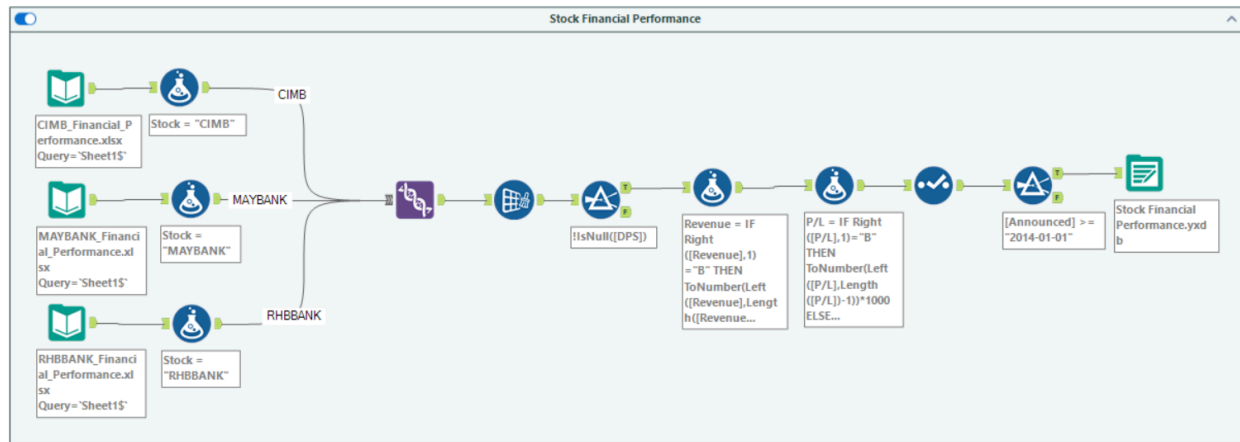


Figure 3 Financial Performance Cleaning and Integration Workflow

Tools	Justification
Input Data	Imported CSV files for financial performance data.
Formula	Added “Stock” column to identify each dataset.
Union	Combined the three financial datasets.
Data Cleanse Pro	Removed null rows and columns from segmented copy-pasted data.
Filter	Retained only rows with financial information.
Formula	Standardised Revenue and P/L to a common unit (M).
Select	Adjusted data types, renamed columns for clarity.
Filter	Retained data from 2014 onwards for forward-fill operations.
Output Data	Saved as Stock Financial Performance.yxdb.

1.3 Economic Indicators

Figure 4 presents the workflow used to prepare the economic indicators dataset. Data sources included the Overnight Policy Rate (OPR) from Bank Negara Malaysia, Gross Domestic Product (GDP) data from OpenDOSM and the Daily Effective Federal Funds Rate (DFF) from FRED.

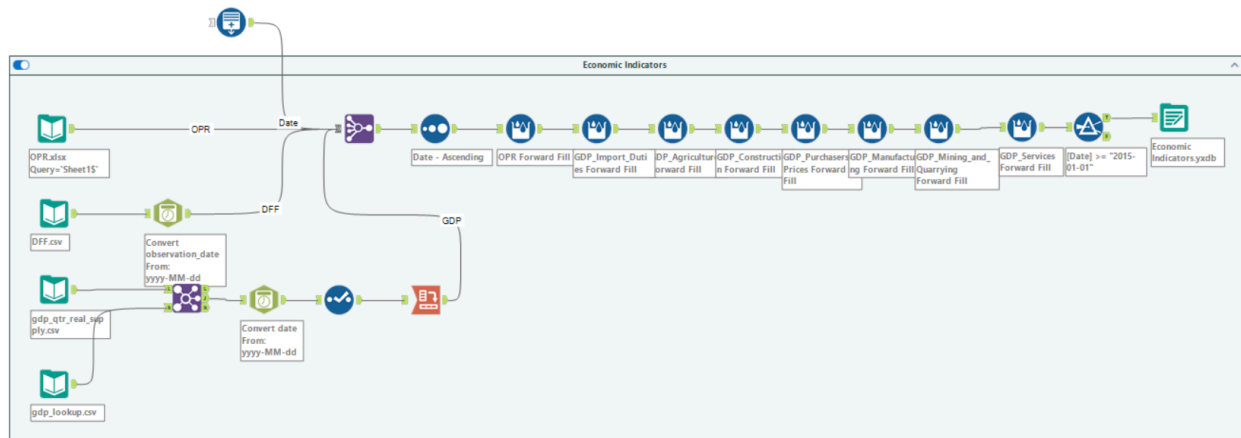


Figure 3 Financial Performance Cleaning and Integration Workflow

Tools	Justification
Input Data	Imported OPR (Bank Negara), GDP (OpenDOSM), DFF (FRED).
Generate Rows	Created continuous date range from 2014-01-01 till 2026-05-29 to facilitate the integration of datasets with different reporting frequencies
Join	GDP sector codes were matched with their corresponding sector names
DateTime	Standardise date fields
Select	Selected relevant columns
Cross Tab	Reshaped the data to create a single record per date with sector columns
Join Multiple	Integrated GDP, OPR and DFF datasets by date.
Sort	The data were sorted by date to facilitate latter forward fill procedures.
Multi-Row Formula	Forward-filled missing values to align different frequencies.
Filter	Retained data from 2015 onwards.
Output Data	Saved as Economic Indicators.yxdb.

The challenges faced in this section was the failure to join the datasets due to the date granularity difference. The problem was solved by adding a continuous date range that allows proper joining of the data.

1.4 Data Integration and Transformation

Figure 5 illustrates the integration of the three prepared datasets and the creation of features.

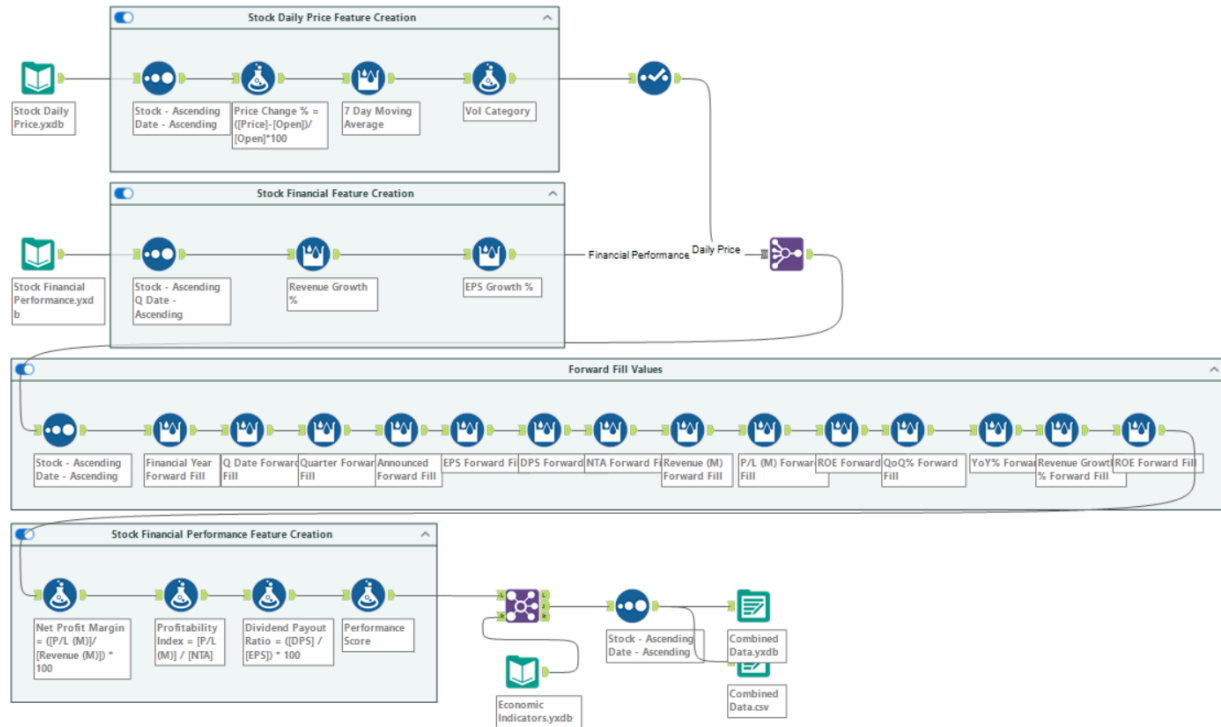


Figure 4 Data Integration and Feature Creation Workflow

Tools	Justification
Input Data	Loaded daily price, financial performance and economic indicator datasets.
Sort, Formula, Multi-Row Formula	Created stock features: Price Change, Volume Category, 7-day MA.
Sort, Multi-Row Formula	Created financial features: Revenue Growth %, EPS Growth %.
Join Multiple	Joined stock price with financial performance by announcement date
Multi-Row Formula	Forward-filled missing values to align different frequencies.
Formula	Calculated Net Profit Margin, Profitability Index, Dividend Payout Ratio and Performance Score.
Join	Integrated stock dataset with economic indicators on date
Sort & Output	Saved final dataset as Combined Data.yxdb and Combined Data.csv.

2. Data Analysis

2.1 Segmentation Analysis: Stock Clustering Using Financial Ratios

Figure 5 shows three financial indicators constructed from the Stock Financial Performance dataset: Earnings Quality (ROE and EPS), Growth Score (YoY% and QoQ% growth) and Income Score (dividend payout ratio). A weighted Cluster Score was then calculated. Based on the weighted Cluster Score, stocks were classified into four categories: Star Performer (strongest overall financial performance), Growth Play (high growth potential with solid fundamentals), Income Stock (stable dividend-oriented performance) and Value Trap (weaker financial performance and fundamentals).

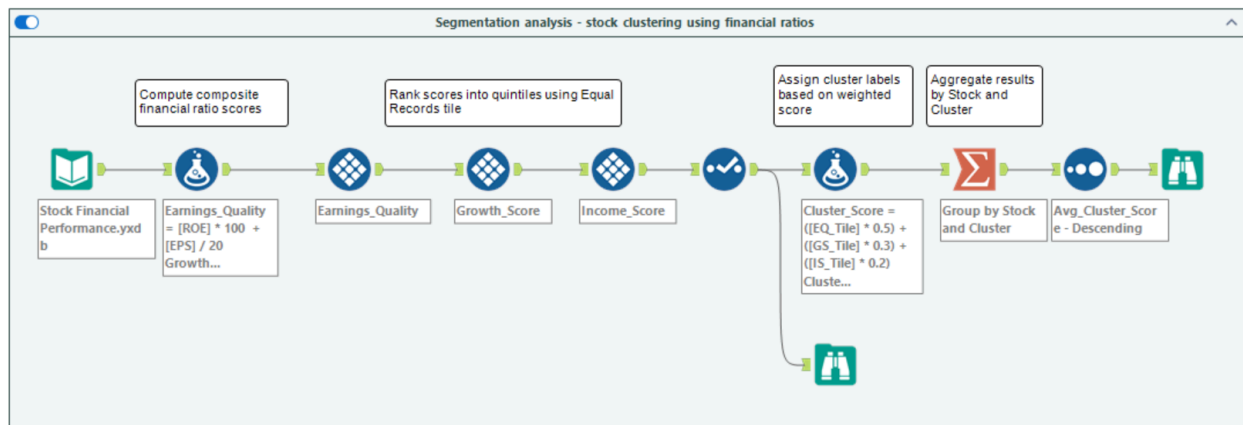


Figure 5 Segmentation Analysis Workflow

Tools	Justification
Input Data	Load Stock Financial Performance
Formula	Compute 3 ratio scores
Tile	Rank each score into quintiles 1–5
Select	Rename Tile_Num → EQ_Tile, GS_Tile, IS_Tile
Formula	Compute Cluster_Score + assign Cluster label
Summarize	Aggregate by Stock + Cluster
Sort	Order by Avg_Cluster_Score descending

Figure 6 showed that none of the banks reached the Star Performer category. Growth Play was the highest cluster achieved, with MAYBANK and RHBBANK appearing more frequently than CIMB. Most observations were classified as Income Stock or Value Trap, with CIMB recording the largest number of Value Trap quarters and the lowest average Cluster Score (1.48).

MAYBANK exhibited the strongest financial performance, achieving the highest average EPS (19.58) and revenue (MYR 14,167 million), while RHBBANK showed the highest average YoY growth (1.32), indicating stronger recovery momentum during weaker periods.

Record	Stock	Cluster	Record_Count	Avg_EPS	Avg_ROE	Avg_YoY%	Avg_Revenue (M)	Avg_Cluster_Score
1	RHBBANK	Growth Play	6	16.683333	0.024167	0.038	3116.666667	3
2	MAYBANK	Growth Play	6	19.583333	0.028	0.047333	14166.666667	3
3	CIMB	Growth Play	1	18.39	0.029	0.019	5500	3
4	MAYBANK	Income Stock	22	18.571364	0.026955	0.084864	13218.181818	2.4
5	RHBBANK	Income Stock	21	16.814286	0.024667	0.117381	3500	2.366667
6	CIMB	Income Stock	18	14.182778	0.026056	0.092444	4550	2.316667
7	MAYBANK	Value Trap	21	17.702857	0.026238	0.263524	11738.095238	1.547619
8	RHBBANK	Value Trap	22	14.248182	0.021545	0.380955	3090.909091	1.536364
9	CIMB	Value Trap	30	10.876667	0.0195	1.318367	4563.333333	1.483333

Figure 6 Segmentation Result

A few challenges were encountered during the analysis. The Tile tool produced a warning due to repeated values within some stock groups, but this did not affect the analysis. Additionally, the Tile tool automatically created generic output names (Tile_Num), which were subsequently renamed using the Select tool to improve clarity and consistency in the workflow.

2.2 Diagnostic Analytics: Driver Analysis of Stock Returns

The Stock Daily Price, Economic Indicators and Stock Financial Performance datasets were integrated using Join tools based on Date and Quarter Key. The explanatory variables were prepared for analysis: OPR_Change, GDP_Svc_Growth, Volume_Surge, ROE, EPS and YoY%. Pearson Correlation was then applied to evaluate the relationship between these variables and Daily_Return, with results ranked by the absolute correlation coefficient (Abs_r).

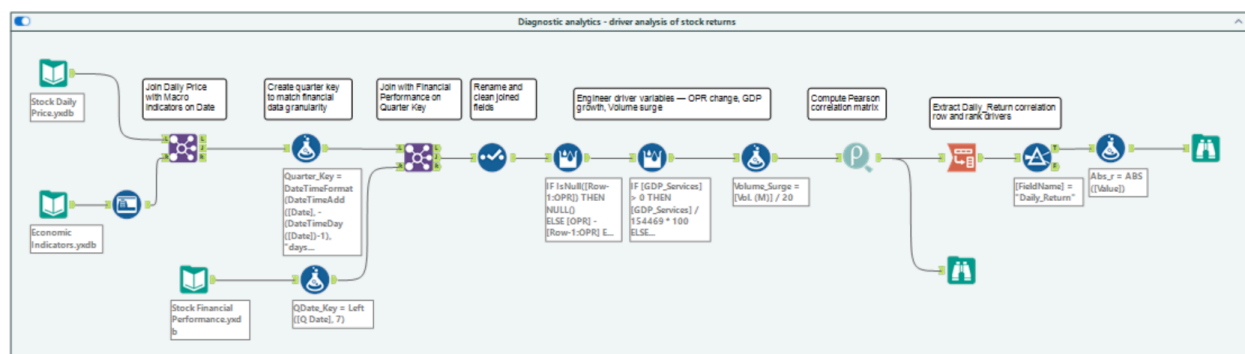


Figure 7 Diagnostic Analytics

Tools	Justification
Input Data	Load Stock Daily Price, Economic Indicators, Stock Financial

	Performance
Auto Field	Converts GDP and DFF fields from String to numeric Double
Join	Join Price+Macro on Date, then join Financial on Quarter Key
Formula	Create Quarter_Key, QDate_Key, OPR_Change, GDP_Svc_Growth, Volume_Surge
Select	Rename Change% → Daily_Return, remove duplicate fields
Multi-Row Formula	Compute OPR_Change row-by-row
Pearson Correlation	Generate correlation matrix of all driver fields
Transpose	Convert wide matrix to long format
Filter	Keep only Daily_Return row from transposed output
Formula	Compute Abs_r = ABS([Value])
Sort	Rank drivers by Abs_r descending

Figure 8 showed that all variables had very weak correlations with Daily_Return ($Abs_r < 0.04$), indicating limited influence on daily stock price movements. OPR_Change recorded the strongest correlation ($r = -0.031$), suggesting that interest rate increases tend to reduce daily returns. Volume_Surge ranked second ($r = -0.028$), implying that higher trading activity was more often associated with selling pressure. Financial indicators such as ROE, EPS and YoY% exhibited near-zero correlations, while GDP_Svc_Growth showed the weakest relationship, indicating minimal short-term impact on daily returns.

Record	FieldName	Name	Value	Abs_r
1	Daily_Return	Daily_Return	1	1
2	Daily_Return	EPS	0.00035	0.00035
3	Daily_Return	ROE	-0.002114	0.002114
4	Daily_Return	YoY%	-0.001802	0.001802
5	Daily_Return	OPR_Change	-0.031293	0.031293
6	Daily_Return	GDP_Svc_Growth	0.000115	0.000115
7	Daily_Return	Volume_Surge	-0.027932	0.027932

Figure 8 Diagnostic Result

Quarter_Key initially returned null values because it was assigned a Date data type instead of V_WString. GDP_Services was stored as a String field and was converted to a numeric format using the Auto Field tool. In addition, the Pearson Correlation output was generated in a wide format, requiring a Transpose tool to extract and analyse the correlation values.

2.3 Time Series Forecasting: Predicting Future Stock Price

Figure 9 represents the multi-stock time series forecasting analysis across independent banking streams. Preliminary data investigation was conducted using TS Plot tools to inspect univariate patterns and visual trend-seasonality decompositions. Each stock stream is then partitioned into training and holdout data to test alternative modeling frameworks, ETS and ARIMA. The superior predictive model was deployed through the TS Forecast tool to generate a 12-month future-horizon price projection.

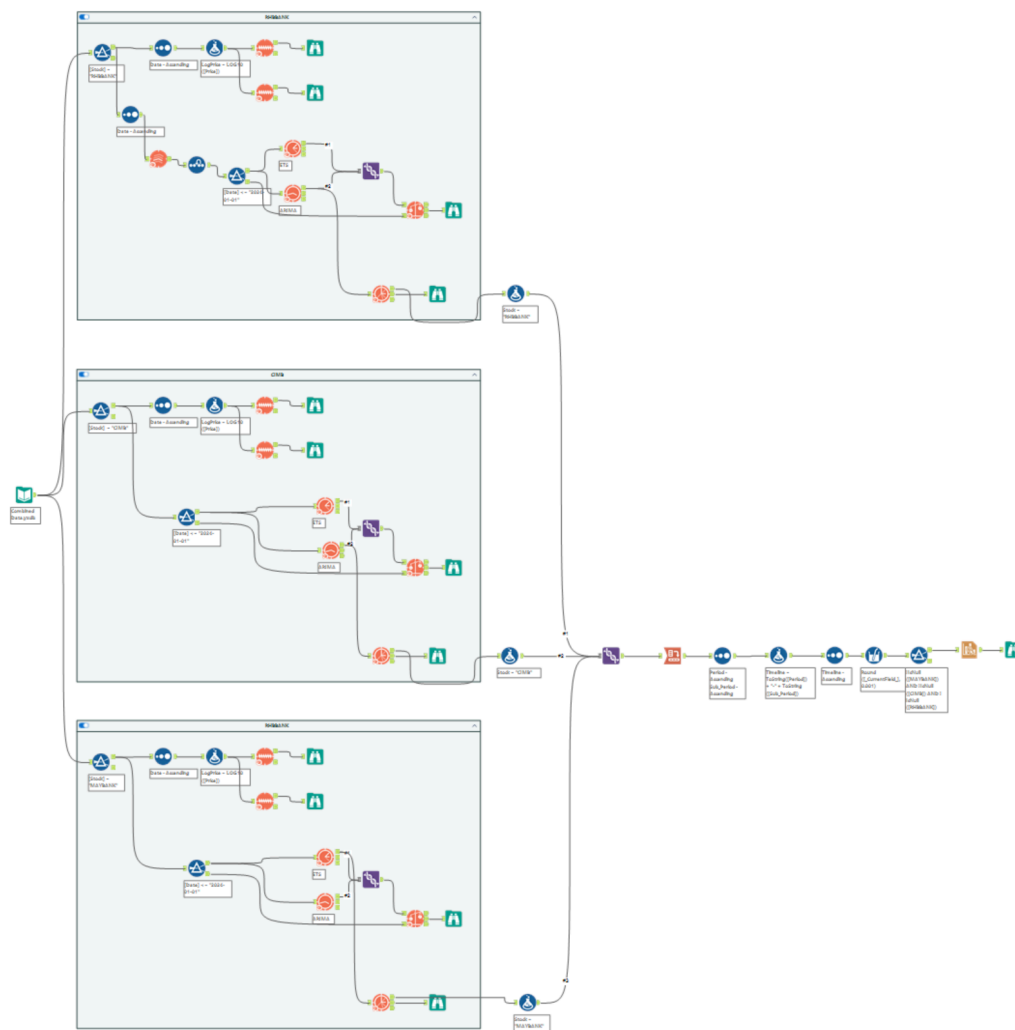


Figure 9 Time Series Forecasting Workflow

Tools	Justification
Input Data	Load Daily Stock Price
Filter	Split data into individual banks
Sort	Orders data by date
Formula	Applies Log10 transformation to stabilize price variance and normalize data distribution
TS Plot	Generate Time Series Plot and Time Series Decomposition Plot by month
Browse	View TS Plots
Filter	Split data into training data and hold data
ETS	Run Experimental Forecast model
ARIMA	Run Experimental Forecast model
Union	Stack experiment results from both ETS and ARIMA model
TS Compare	Compare experimental forecast result with hold data
Browse	View model comparison result
TS Forecast	Generates future point forecast using chosen model
Browse	View forecast result
Formula	Apply tag to identify stock
Union	Stack forecast output of three stock together
Cross Tab	Reshape data, transform stock name from vertical to horizontal
Sort	Align merged prediction result by period and sub-period
Formula	Concatenate numerical period identifiers into string column, timeline
Sort	Sort data by timeline
Multi-Field Formula	Round off predicted stock price into 3 decimal places
Filter	Drop Null row
Interactive Chart	Generate 3-month multi-line visual comparison graph

Browse	View Chart
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The forecast result shows CIMB as the optimal top growth stock with volatility, while Maybank and RHB stock are highly predictable, low-risk defensive with flatline forecasts indicating minimal price drift over the 2027 horizon.

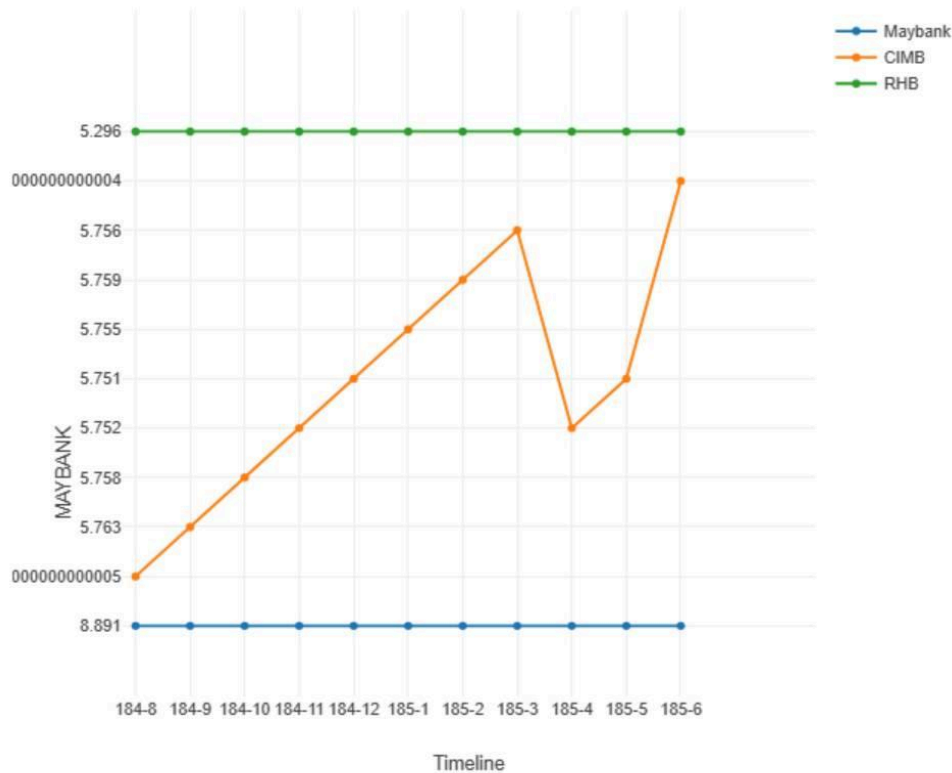


Figure 10 Time Series Forecasting Comparison Chart

The primary challenge arose because time series models track timelines sequentially by row counts rather than calendar dates, and RHB Bank stock price record is 21 records shorter than others, causing the final forecast between the banking streams to be misaligned and fragmented. This problem is resolved by using a Time Series Filler tool to automatically scan and pad the missing calendar gaps for each RHB bank stream. An Imputation tool was then deployed immediately downstream to replace the resulting null values with statistical averages, to standardize the row counts and ensure a synchronized final timeline comparison.